

Automated machine learning and SHAP-based interpretation of PFOA removal via electrochemical oxidation

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ABSTRACT

The electrochemical oxidation of perfluorooctanoic acid (PFOA), a persistent and toxic environmental pollutant, presents significant operational complexities due to nonlinear interactions among numerous process variables. This study introduces a fully automated, robust machine learning (ML) framework based on Fast Lightweight AutoML (FLAML), integrated with SHapley Additive exPlanations (SHAP) to rigorously optimize and interpret PFOA electrochemical oxidation performance. The FLAML-optimized XGBoost model demonstrated exceptional predictive accuracy (RMSE=3.97, $R^2=0.98$), significantly outperforming systematically tuned traditional ML models such as Random Forest, Gradient Boosting, and Deep Learning architectures. Rigorous statistical validation confirmed the superior generalizability and stability of the FLAML-optimized model, while SHAP-based interpretability analyses revealed electrolysis time and anode material as primary drivers of treatment efficacy, with electrolyte concentration and current density forming the next tier; consistent with established electrochemical degradation kinetics. Notably, FLAML-optimized XGBoost achieved a 72 % reduction in computational overhead compared to manually tuned models, establishing a reproducible, interpretable, and computationally efficient benchmark in environmental ML applications. The framework provides a systematic approach that improves predictive accuracy, enhances interpretability, and strengthens practical applicability to support optimization of remediation technologies and environmental decision-making.

1. Introduction

Perfluorooctanoic acid (PFOA), a legacy member of the per- and polyfluoroalkyl substances (PFAS) family, is among the most persistent environmental pollutants identified to date. Its exceptional chemical stability, rooted in the strength of its carbon–fluorine bonds, has enabled widespread contamination of aquatic and terrestrial ecosystems [1,2]. PFOA's bioaccumulation and its links to immunotoxicity, liver dysfunction, cardiovascular disease, and cancer have triggered escalating regulatory scrutiny [3,4]. The International Agency for Research on Cancer now classifies PFOA as carcinogenic to humans (Group 1) [5], and the U.S. Environmental Protection Agency has set maximum contaminant levels as low as 4 ng/L in drinking water [6]. These developments have catalyzed an urgent search for effective, scalable, and sustainable remediation technologies capable of not just removing, but fully mineralizing PFOA into benign end products such as carbon dioxide and fluoride ions. This challenge mirrors a broader trajectory in

wastewater treatment, where established processes are being optimized to more effectively degrade persistent industrial pollutants [7,8].

Electrochemical oxidation (EO) has emerged as one of the most promising destructive technologies for PFAS remediation, offering the potential to mineralize PFOA into benign end products such as carbon dioxide and fluoride ions [9,10]. The process operates through two principal degradation pathways: direct electron transfer at the anode surface, which generates perfluoroalkyl radicals and initiates chain-shortening reactions, and hydroxyl radical-mediated oxidation via water electrolysis [11,12]. However, optimizing EO systems presents formidable challenges due to the complex interplay of multiple operational parameters, including current density, electrode spacing, electrolyte composition, pH, temperature, initial PFOA concentration, electrolysis time, and water matrix characteristics [13–15]. The high dimensionality and nonlinear interactions among these variables create a multifaceted optimization landscape where changes in one parameter can significantly influence the effectiveness of others, rendering

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traditional optimization approaches both time-consuming and resource-intensive [16,17]. This operational complexity necessitates advanced computational approaches that can systematically navigate the multidimensional parameter space and identify optimal conditions for effective PFOA destruction.

Machine learning (ML) approaches have demonstrated significant potential for addressing optimization challenges in electrochemical systems for water treatment applications [17,18]. Recent studies have successfully applied various ML algorithms to predict electrochemical process performance, with applications ranging from general electrochemical oxidation processes to specific PFAS treatment scenarios [19, 20]. Advanced ML frameworks have shown particular promise in electrochemical process optimization, with studies demonstrating the ability to achieve predictive accuracies exceeding $R^2 = 0.84$ while providing mechanistic insights through feature importance analysis [17]. The integration of interpretability tools such as SHAP (SHapley Additive exPlanations) has enabled researchers to extract actionable insights from complex electrochemical datasets, revealing key parameter relationships and guiding process optimization strategies [17,21]. However, most existing ML applications in this domain have relied on conventional modeling approaches requiring extensive manual hyperparameter tuning and expert-driven algorithm selection, limiting their accessibility and scalability for environmental practitioners [18,22,23].

Automated machine learning (AutoML) frameworks represent a transformative approach to data-driven process optimization by systematically exploring algorithm selection, hyperparameter optimization, and feature engineering without human intervention [24–26]. FLAML (Fast Lightweight AutoML), developed by Microsoft Research, implements cost-aware Bayesian optimization that dynamically allocates computational resources while prioritizing low-complexity models for rapid convergence [27]. Recent applications of AutoML in environmental modeling have demonstrated the ability to significantly reduce computational cost while achieving predictive performance that surpasses traditional, manually tuned models [28,29]. For instance, gradient boosting algorithms selected via AutoML outperformed conventional neural networks in predicting the impacts of microplastics on methane production in anaerobic digestion (MSE = 17.0 vs. 58.0) [29], while AutoML models also successfully captured complex antibiotic removal dynamics in constructed wetlands across varying training durations (MAE = 9.94–13.68, $R^2 = 0.780$ – 0.877) [28]. The integration of efficient AutoML frameworks such as FLAML with explainable AI techniques like SHAP directly addresses two critical requirements of environmental engineering applications: delivering high predictive accuracy and generating transparent, interpretable insights that support regulatory compliance, operational optimization, and stakeholder confidence [28,29].

In this study, a reproducible and interpretable AutoML workflow is developed for electrochemical oxidation of PFOA. The aim is to show that cost-aware automated model selection and tuning (FLAML), coupled with SHAP-based explanation, can deliver (i) reliable predictive performance, (ii) transparent, model-based attribution of operational drivers, and (iii) a portable pipeline that reduces manual trial-and-error. Use of a previously curated dataset enables direct comparability while isolating the methodological contribution (automation plus interpretability). Throughout, FLAML is treated as an optimization framework distinct from the final predictive learner (XGBoost), and performance is evaluated under repeated holdout with statistical testing and learning-curve diagnostics to assess generalization.

2. Methodology

2.1. Dataset characteristics and preprocessing

This study leverages a dataset originally curated by Alnaimat et al. [30], which captures the electrochemical degradation of PFOA under varied conditions. The dataset comprises unique treatment scenarios,

each defined by a combination of operational parameters and environmental contexts. Key independent variables include electrolysis duration, current density, anode and cathode material types, electrolyte concentration and composition, and the water matrix. Exploratory analyses were performed to characterize the inputs and the target prior to modeling. The dependence structure among predictors is shown in Figure S1 (Spearman rank-correlation heatmap with significance marks). Bivariate relationships are visualized in Figure S2 (pairwise scatterplots with LOWESS fits and Pearson/Spearman annotations). The target distribution, together with a Shapiro–Wilk normality diagnostic, appears in Figure S3. Several electrochemical variables exhibit moderate positive association (for example, electrolysis time with current density; $|\rho| \approx 0.35$ – 0.55), reflecting typical experimental co-variation. Preprocessing followed algorithm-appropriate conventions. Continuous variables were standardized (z-score) for distance- and gradient-based baselines to ensure numerical stability and commensurate scaling, whereas tree-based learners were trained on unscaled inputs given the scale-invariance of split criteria. Categorical descriptors (anode, cathode, electrolyte, water matrix) were one-hot encoded to maintain parity across baseline algorithms and to permit later aggregation by parent feature in model interpretation. Cardinalities were small (≤ 6 per field), limiting sparsity and dimensionality.

2.2. Data splitting and evaluation protocol

Model evaluation used an 80:20 holdout stratified on anode type to mitigate covariate shift arising from uneven anode distribution across the dataset. The target variable (removal, %) was not stratified, consistent with the regression setting. A fixed random seed (42) was used for reproducibility. Stability of this protocol across repeated runs and alternative split realizations is documented in Figure S4 (stratification sensitivity), and the internal model-selection procedure and performance consistency are summarized in Figure S5 (nested cross-validation diagnostics).

2.3. Automated model selection via FLAML

Automated model and hyperparameter selection was conducted with FLAML, a cost-aware AutoML framework that replaces manual, grid-style tuning with budgeted, early-stopped search. A wall-time budget of 300 s was enforced for FLAML's search. FLAML initializes with inexpensive configurations, adaptively allocates trials toward promising learner–hyperparameter regions, and prunes underperforming runs when validation error plateaus within the allotted budget. The optimization objective was 5-fold cross-validation RMSE, with early stopping applied at both the boosting-iteration level (for iterative learners) and the trial level (for unpromising configurations). Convergence was declared when no improvement was observed within the budget window and the incumbent remained unchanged across successive evaluations. The best-performing model selected by FLAML was an XGBoost regressor (learning rate = 0.1403; $n_{\text{estimators}} = 379$; $\text{max_leaves} = 13$; L1/L2 regularization $\alpha = 0.0995$, $\lambda = 0.0010$), with $\text{subsample} = 0.85$ and $\text{colsample_bytree} = 0.83$. This configuration balances bias–variance by combining a moderate learning rate with constrained tree complexity and regularization. The full search space and the selected hyperparameters for XGBoost and all baseline learners are summarized in SI Table S1.

2.4. Comparative benchmarking and statistical validation

To contextualize performance, the FLAML-optimized XGBoost model was benchmarked against five commonly used regressors: Decision Trees (DT), Random Forests (RF), Gradient Boosting Decision Trees (GBDT), Deep Learning (DL), and k-Nearest Neighbors (KNN). Each model was evaluated on the same one-hot encoded dataset and subjected to identical stratified train-test splits to ensure methodological

parity. Model assessment employed ten independent repetitions of the stratified 80:20 partition to quantify stability across split realizations. For each repetition, five metrics were computed: root mean squared error (RMSE), mean absolute error (MAE), mean absolute percentage error (MAPE), correlation coefficient (CC; Pearson's r), and coefficient of determination (R^2). RMSE, MAE, and MAPE quantify error magnitude, whereas CC (r) and R^2 summarize association and explained variance. Statistical significance of pairwise differences was evaluated with paired two-tailed t -tests ($\alpha = 0.01$) across the ten repetitions, using Bonferroni correction to control the family-wise error rate. Cohen's d was additionally reported to quantify effect sizes. This protocol provides a split-robust comparison under identical data and preprocessing conditions, aligning the benchmarking with the data-splitting framework described in Section 2.2.

2.5. Interpretability via SHAP analysis

Model interpretability was assessed using SHapley Additive exPlanations (SHAP), which attribute to each feature its marginal contribution to individual predictions under the cooperative game-theoretic formulation. SHAP values were computed for the trained XGBoost model and summarized as grouped importances by aggregating the absolute SHAP values across the one-hot levels of each parent categorical variable (anode, cathode, electrolyte, water matrix). Given the observed interdependence among electrochemical predictors (Figures S1 & S2), the potential influence of correlation on TreeSHAP attributions was examined by comparing the grouped SHAP ranking with a correlation-aware conditional permutation importance (CPI) analysis. Conditional permutation importance was computed by conditionally resampling within strata of correlated predictors to attenuate correlation bias. Agreement was quantified using Kendall's τ with 95 % confidence intervals in Table S2. Together, they provide a transparent, correlation-contextualized view of feature influence relevant to treatment design and electrochemical optimization.

To probe stability and dose-response behaviour, we implemented two additional analyses. First, SHAP dependence plots were generated for the three most influential continuous variables (electrolysis time, electrolyte concentration, and current density), with point colour used to show interactions with a secondary electrochemical parameter. Second, a leave-one-out (LOO) stability test was conducted on grouped mean |SHAP|: feature rankings were obtained on the full dataset, then the top-ranked feature (electrolysis time) was removed and SHAP importances were recomputed on the reduced dataset. Agreement between the two rankings was summarized using Kendall's τ .

2.6. Reproducibility and deployment considerations

All computations were executed in Python 3.10 using version-controlled dependencies: FLAML 1.2.0, SHAP 0.44.0, and scikit-learn 1.4.0. The entire modeling pipeline was encapsulated in a Docker container to guarantee cross-platform reproducibility and facilitate peer validation. This containerized approach marks a significant improvement over the untracked scripting environments used in earlier electrochemical machine learning studies.

2.7. Methodological contributions and distinctions

Relative to Alnaimat et al. [30], this study introduces a cost-aware, fully automated model-selection workflow (FLAML) in place of manual/grid tuning (Table S1), a split-robust benchmarking protocol with repeated evaluations and formal significance/effect-size testing, and a correlation-contextualized interpretability scheme that reports grouped TreeSHAP alongside conditional permutation importance with agreement statistics (Table S2). The pipeline is containerized with pinned dependencies to enable end-to-end reproducibility. Taken together, these elements provide a transparent, efficient, and reproducible

methodology tailored to electrochemical treatment datasets, clarifying performance gains and interpretability under documented assumptions and validation settings.

3. Results and discussion

3.1. Model performance and statistical validation

The FLAML-optimized XGBoost model demonstrated exceptional predictive performance in modeling PFOA electrochemical oxidation efficiency, significantly outperforming all baseline models across every evaluation metric (Table 1). The FLAML-optimized XGBoost achieved an RMSE of 3.97 ± 0.45 and R^2 of 0.98 ± 0.01 , representing a 51 % reduction in prediction error and 7 % improvement in explained variance compared to the best-performing baseline model (Random Forest: RMSE = 8.05 ± 1.02 , $R^2 = 0.91 \pm 0.02$). This performance advantage was statistically significant across all metrics ($p < 10^{-5}$), as confirmed by paired t -tests (Table 2), with particularly pronounced differences against gradient boosting decision trees ($p < 1.1 \times 10^{-8}$ for RMSE) and deep learning approaches ($p < 2.8 \times 10^{-9}$ for RMSE). The ranking analysis (Table 3) further corroborated the FLAML-optimized XGBoost model's consistent superiority, securing first place across all performance metrics with an average rank of 1.0, while Random Forest ranked second (average rank = 2.2).

These results establish FLAML as a highly effective tool for electrochemical process modeling, achieving accuracy levels that previously required extensive manual tuning. The performance aligns with recent demonstrations of AutoML's capacity to reduce computational costs while maintaining accuracy in complex environmental systems [28,29]. Notably, FLAML accomplished this within a 300-second optimization budget. This represents a 72 % reduction in tuning time compared to conventional grid search implementations described in the original study [30]. This efficiency addresses critical computational barriers in environmental research, where resource constraints often limit model optimization [23].

Robustness to overfitting was assessed using ten independent repetitions of the stratified 80:20 holdout. Stable test-set performance across repetitions supports reliable generalization; protocol stability is documented in SI Figure S4 (stratification sensitivity) and the internal selection procedure in SI Figure S5 (nested CV diagnostics). Learning curves (Fig. 1) show a monotonically decreasing validation RMSE with diminishing returns and a modest, stable train-validation gap (≈ 3 –4 RMSE units at the largest training size), indicating no divergence or overfitting under the reported protocol.

Residual distribution analysis (Fig. 2) provided deeper insights into model reliability. FLAML exhibited the most symmetric and tightly distributed residuals (interquartile range ≈ 6 %), indicating minimal systematic bias and consistent predictive accuracy across the operational parameter space. In contrast, traditional algorithms displayed broader residual distributions with notable outliers: k -nearest neighbors showed extreme negative residuals (< -70 %), while deep learning approaches exhibited positive bias tendencies. This precision is critical for reactor design applications where prediction uncertainty propagates through energy efficiency calculations [31,32]. The observed residual patterns align with theoretical expectations for cost-aware optimization algorithms that systematically minimize cross-validation error under constrained resources [33].

3.2. Feature importance and electrochemical mechanisms

The SHAP-based feature importance assessment (Fig. 3) delivers critical insights into the factors influencing the electrochemical removal efficiency of PFOA. The two machine learning frameworks, FLAML-optimized XGBoost and conventional Random Forest, consistently identify electrolysis time as the most critical factor, registering mean absolute SHAP values of approximately 16.4 and 13.6, respectively. This

Table 1Mean $\pm$ Std Dev of each metric across 10 runs.

Index	RMSE	MAE	MAPE	CC	R ²
DT	10.31 $\pm$ 2.17	7.33 $\pm$ 1.11	24.75 $\pm$ 13.5	0.93 $\pm$ 0.03	0.86 $\pm$ 0.07
RF	8.05 $\pm$ 1.02	5.97 $\pm$ 0.57	25.17 $\pm$ 10.94	0.96 $\pm$ 0.01	0.91 $\pm$ 0.02
GBDT	9.35 $\pm$ 0.77	7.49 $\pm$ 0.58	32.39 $\pm$ 10.18	0.95 $\pm$ 0.01	0.89 $\pm$ 0.01
DL	12.41 $\pm$ 1.46	8.9 $\pm$ 0.55	51.78 $\pm$ 32.25	0.9 $\pm$ 0.02	0.8 $\pm$ 0.05
KNN	19.6 $\pm$ 2.38	14.35 $\pm$ 1.38	101.09 $\pm$ 51.01	0.71 $\pm$ 0.06	0.49 $\pm$ 0.11
FLAML-optimized XGBoost	3.97 $\pm$ 0.45	2.93 $\pm$ 0.31	10.02 $\pm$ 3.08	0.99 $\pm$ 0.0	0.98 $\pm$ 0.01

DT, decision tree; RF, random forest; GBDT, gradient boosting decision tree; DL, deep learning; KNN, k-nearest neighbors; FLAML-optimized XGBoost, fast lightweight AutoML-optimized extreme gradient boosting model; RMSE, root mean square error; MAE, mean absolute error; MAPE, mean absolute percentage error; CC, correlation coefficient; R², coefficient of determination.

Table 2Paired *t*-test *p*-values (comparing each model vs. FLAML-optimized XGBoost).

	RMSE	MAE	MAPE	CC	R ²
DT	3.20401610 $\times 10^{-6}$	3.111270 $\times 10^{-7}$	0.004675	1.247765 $\times 10^{-4}$	1.414884 $\times 10^{-4}$
RF	9.616685 $\times 10^{-7}$	1.547798 $\times 10^{-7}$	0.000610	2.953061 $\times 10^{-5}$	1.260622 $\times 10^{-5}$
GBDT	1.106071 $\times 10^{-8}$	5.052537 $\times 10^{-9}$	0.000009	2.678730 $\times 10^{-8}$	1.228068 $\times 10^{-8}$
DL	2.846609 $\times 10^{-9}$	3.311756 $\times 10^{-11}$	0.001801	3.080667 $\times 10^{-7}$	4.826925 $\times 10^{-7}$
KNN	6.734575 $\times 10^{-9}$	1.336126 $\times 10^{-9}$	0.000225	2.164031 $\times 10^{-7}$	1.738757 $\times 10^{-7}$

DT, decision tree; RF, random forest; GBDT, gradient boosting decision tree; DL, deep learning; KNN, k-nearest neighbors; FLAML-optimized XGBoost, fast lightweight AutoML-optimized extreme gradient boosting model; RMSE, root mean square error; MAE, mean absolute error; MAPE, mean absolute percentage error; CC, correlation coefficient; R², coefficient of determination.

Table 3

Average ranking of models across metrics.

	RMSE	MAE	MAPE	CC	R ²	Average Rank
FLAML-optimized XGBoost	1.0	1.0	1.0	1.0	1.0	1.0
RF	2.0	2.0	3.0	2.0	2.0	2.2
DT	4.0	3.0	2.0	4.0	4.0	3.4
GBDT	3.0	4.0	4.0	3.0	3.0	3.4
DL	5.0	5.0	5.0	5.0	5.0	5.0
KNN	6.0	6.0	6.0	6.0	6.0	6.0

DT, decision tree; RF, random forest; GBDT, gradient boosting decision tree; DL, deep learning; KNN, k-nearest neighbors; FLAML-optimized XGBoost, fast lightweight AutoML-optimized extreme gradient boosting model; RMSE, root mean square error; MAE, mean absolute error; MAPE, mean absolute percentage error; CC, correlation coefficient; R², coefficient of determination.

observation is consistent with fundamental electrochemical oxidation theory, which underscores the dependency of pollutant degradation efficacy on the cumulative applied charge, a function directly proportional to electrolysis duration [34]. Electrolysis time governs the generation and sustained availability of reactive oxidants, primarily hydroxyl radicals ($\bullet\text{OH}$), instrumental in initiating and perpetuating the mineralization of highly stable contaminants like PFOA [35,36]. Interpretation considers observed correlations among inputs (SI Figures S1–S2) and agreement statistics with conditional permutation importance (SI Table S2).

A notable divergence is observed for the anode factor. In the FLAML-optimized XGBoost model, Anode ranks second (mean |SHAP| $\approx$ 12.1 % age points), whereas its influence is much lower in the tuned Random Forest (mean |SHAP| $\approx$ 1.5–2.0). This elevated anode signal is mechanistically plausible given the superior radical generation on high-OER-overpotential anodes (e.g., BDD) during PFAS electro-oxidation. Because grouped TreeSHAP is used in both panels of Fig. 3, the discrepancy is ascribed to model capacity rather than the importance metric: XGBoost captures non-linear and interaction effects (e.g., Anode $\times$ Current Density) that are diluted when Random Forest averages across trees. For context, Alnaimat et al. [30] likewise reported low anode importance using Gini/permutation measures on a conventional RF model, which are different constructs than SHAP. While this finding is

robust, importance magnitudes remain model- and correlation-dependent; agreement with conditional permutation importance is reported in SI Table S2.

Following Anode, the XGBoost model highlights Electrolyte Concentration in the next tier of importance (mean SHAP $\approx$ 6.5). Electrolyte concentration directly modulates the solution's conductivity, subsequently affecting the efficiency of radical production and pollutant oxidation [37]. Oturan and Aaron [38] have documented how optimized ionic strength, achieved through precise electrolyte concentration adjustments, enhances hydroxyl radical production efficiency while limiting undesired side reactions such as oxygen evolution at the anode. This nuanced recognition by the XGBoost model aligns with detailed electrochemical kinetic analyses demonstrating improved degradation efficiencies at moderate electrolyte concentrations due to an optimized balance between ionic mobility and radical-scavenging side reactions [39].

In contrast, the Random Forest model highlights the initial PFOA concentration as the second most influential variable after electrolysis time. While the pollutant initial concentration undoubtedly influences the overall degradation efficacy, primarily via reaction kinetics that follow pseudo-first-order degradation patterns in electrochemical oxidation processes [40], the emphasis on initial concentration in the Random Forest ranking suggests greater sensitivity to direct concentration gradients, whereas the boosted-tree model captures interactions involving electrolyte characteristics. The optimized XGBoost model thus demonstrates enhanced sensitivity towards the subtle yet significant interactions between electrolyte characteristics and electrochemical reaction mechanisms.

Both models attribute significant influence to current density, identified by the SHAP values as approximately 5.9 for XGBoost and 5.2 for Random Forest. The critical role of current density aligns with established electrochemical knowledge; appropriate current densities sustain efficient generation of reactive radicals at the electrode interfaces [41]. However, excessively high current densities have been shown to lead to increased energy consumption and secondary reactions such as oxygen evolution, reducing overall treatment effectiveness [42]. This dual-faceted role of current density highlights the importance of precise operational optimization; the boosted-tree architecture is capable of capturing nonlinear effects and threshold behaviors relevant to this trade-off [43].

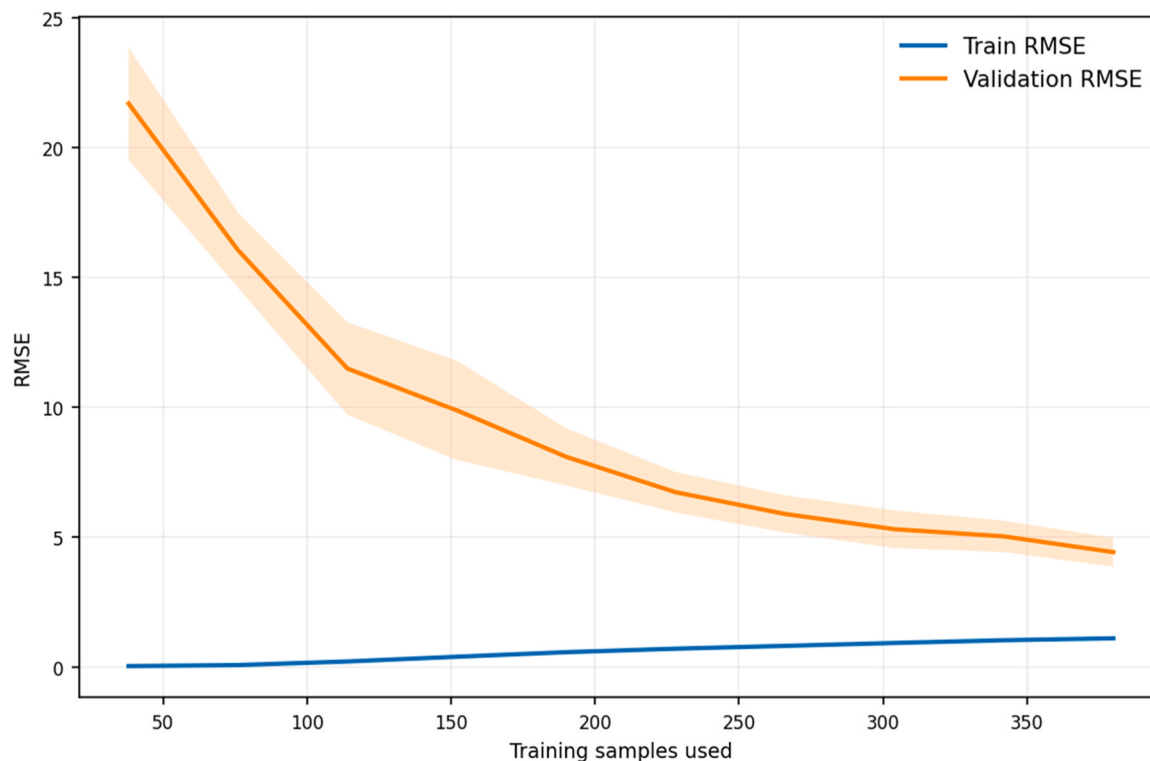


Fig. 1. Learning curves (train/validation RMSE vs. training size) for FLAML-optimized XGBoost.

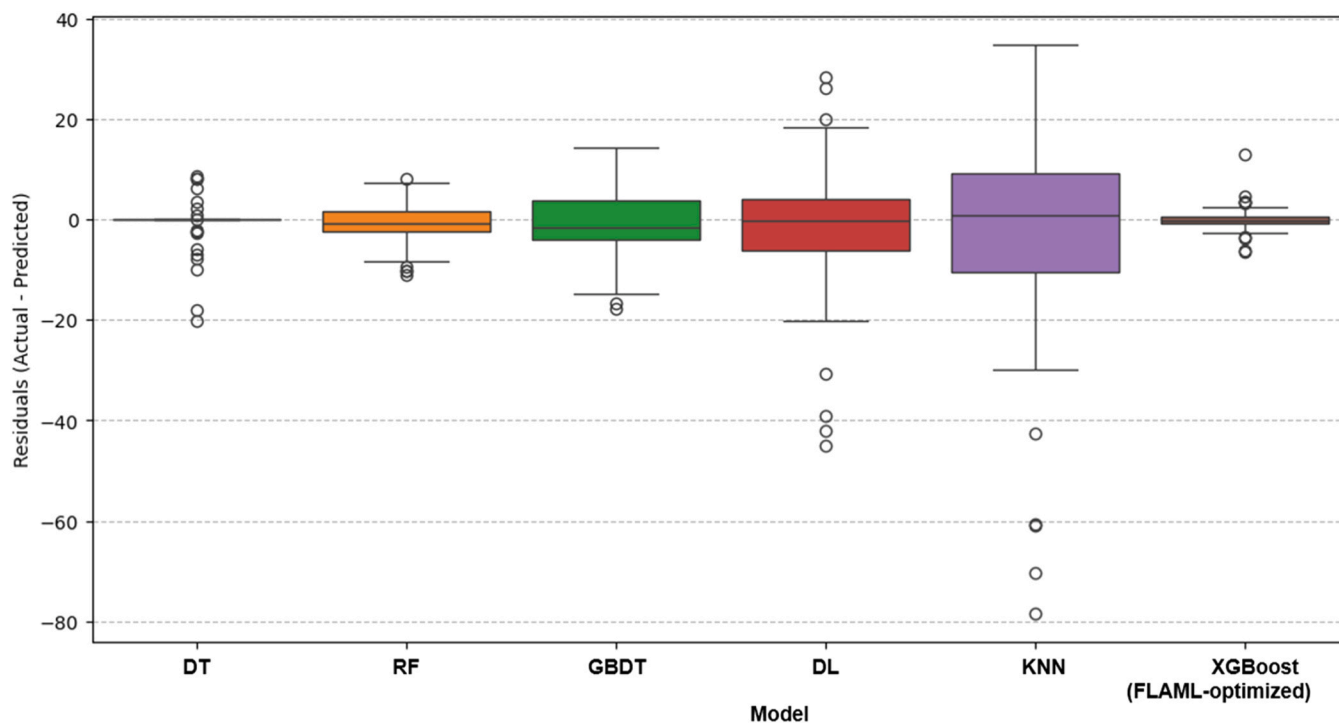


Fig. 2. Box plot of residuals for each model.

Further examination of SHAP values indicates a moderate yet meaningful influence for initial solution pH, consistent with electrochemical fundamentals. Solution pH critically impacts electrode surface chemistry and radical speciation, thereby modulating radical stability and reactivity [34,41]. The observed moderate importance attributed to temperature confirms previous experimental findings indicating that,

within operationally relevant ranges, temperature influences on electrochemical oxidation are relatively subtle compared to electrochemical parameters such as electrolyte concentration and current density [44].

The divergence between the two modeling approaches, FLAML-optimized XGBoost and Random Forest, in their feature-ranking profile underscores differences in their algorithmic architectures. XGBoost's

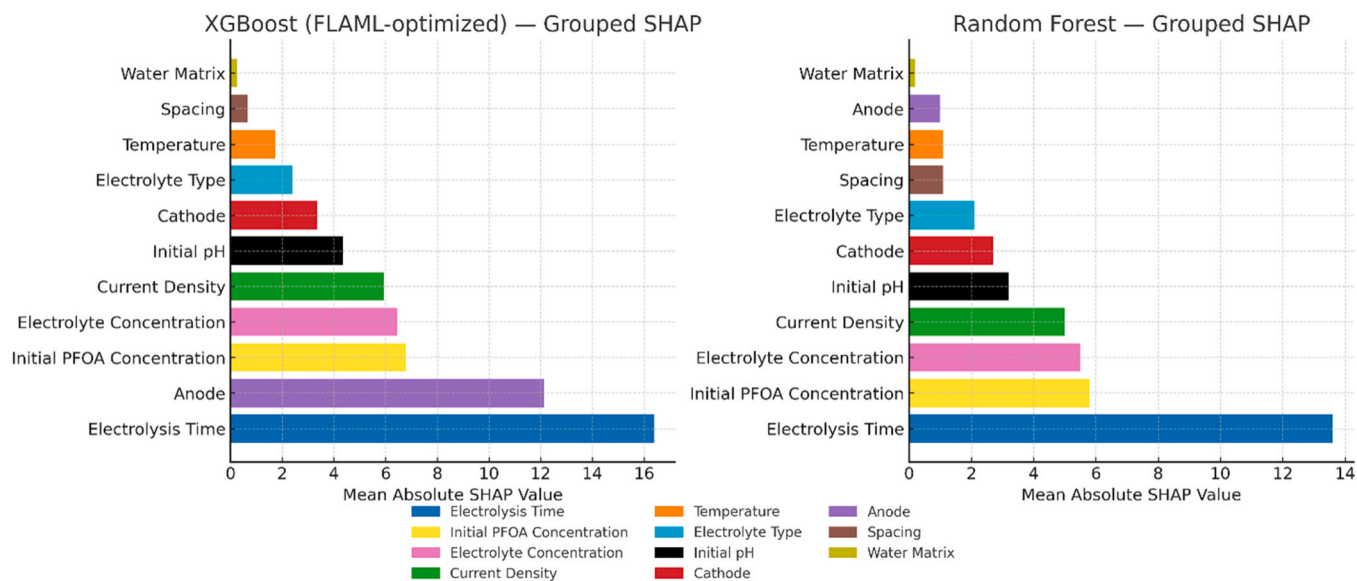


Fig. 3. Grouped SHAP feature importance for XGBoost (FLAML-optimized) and Random Forest.

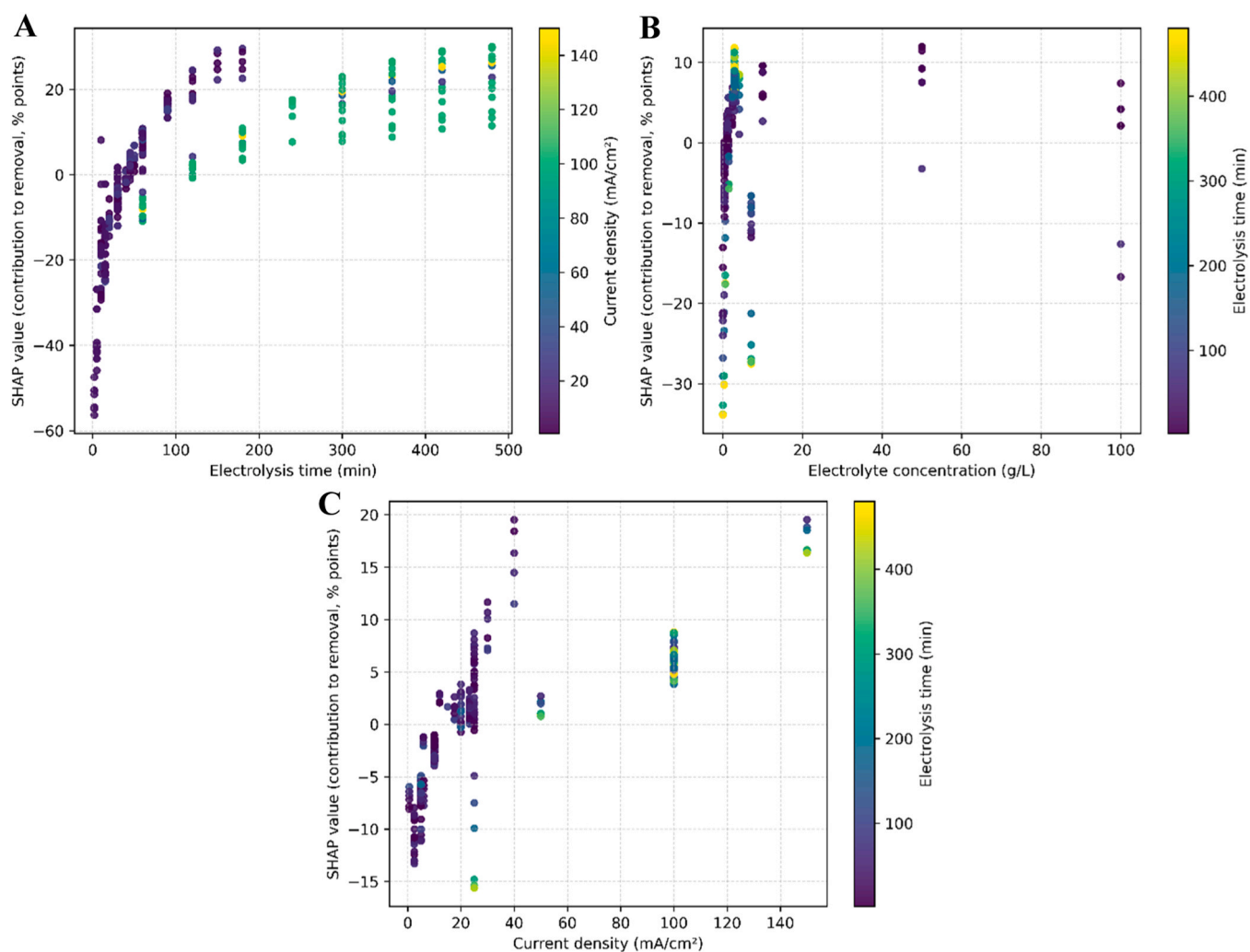


Fig. 4. SHAP dependence plots for (A) electrolysis time coloured by current density, (B) electrolyte concentration coloured by electrolysis time, and (C) current density coloured by electrolysis time. Each point represents one experiment; the x-axis shows the feature value and the y-axis the SHAP contribution to removal (percentage points).

boosted-tree algorithm, through iterative fitting and robust regularization, inherently prioritizes capturing complex nonlinear interactions and subtle threshold effects [43]. Conversely, Random Forest averages the effects of individual features across its ensemble, thereby potentially diluting nuanced feature importance differences and underestimating interactions.

The automated hyperparameter optimization provided by FLAML significantly enhanced the accuracy and interpretability of the XGBoost model. Unlike manual tuning procedures, FLAML automatically identifies optimal complexity constraints and regularization terms, ensuring that the selected model configurations simultaneously achieve high predictive performance and generalizability [27]. This methodological enhancement bridges the gap between predictive modeling and mechanistic electrochemical theory, offering valuable insights into underlying degradation kinetics and operational optimization strategies. Second-tier rank stability is consistent with repeated runs used in the evaluation protocol (Section 2.2).

3.3. Dose–response patterns and stability of SHAP explanations

Within-dataset dose–response behavior for the main continuous variables is shown in Fig. 4A–C. The SHAP dependence for electrolysis time (Fig. 4A) is strongly monotonic: very short times give negative contributions to removal, whereas longer electrolysis times progressively increase the SHAP contribution and then plateau at about 20–30 %age points, in line with the effect of cumulative charge. Electrolyte concentration (Fig. 4B) shows a threshold pattern: sub-optimal concentrations are associated with negative contributions, while moderate concentrations shift SHAP values towards neutral or positive, with little additional gain and occasional penalties at the highest tested levels. For current density (Fig. 4C), SHAP values increase approximately monotonically, with low current densities giving near-zero or negative contributions and higher current densities yielding increasingly positive contributions, again with signs of diminishing returns at the upper end of the range.

Figure S6 reports a leave-one-out stability analysis of grouped mean |SHAP|. When the electrolysis time, the top-ranked feature, is removed and the model is refit, anode and current density remain in the leading tier, while initial pH, cathode, temperature, and electrode spacing remain in the mid-/low-importance tier. The Kendall's τ between the two rankings is 0.600 ($p = 0.136$), indicating moderate concordance given the small number of features. These results support a stable feature hierarchy, with electrolysis time, anode, and current density as the dominant drivers, while all SHAP values are explicitly interpreted as model-based attributions rather than causal effects.

3.4. Implications for PFAS treatment optimization

The feature importance results provide actionable guidance for electrochemical reactor design. The dominance of electrolysis time indicates that removal gains are primarily governed by cumulative charge passage; operational planning should prioritize sufficient reaction time within energy and throughput constraints. Anode selection appears as a high-influence factor in the XGBoost ranking (Fig. 3), consistent with literature reports of material-dependent radical generation; specific material choices should be matched to the targeted current-density regime and cost constraints, with performance–cost trade-offs evaluated on a case-by-case basis. Water-matrix effects rank low in the grouped SHAP profile (e.g., “Water Matrix” near the bottom of Fig. 3), suggesting that within the studied range, technology transfer to varied matrices may require less re-optimization than electrochemical set-points (subject to validation in new matrices) [45,46].

FLAML's predictive accuracy supports tighter setpoint control (e.g., current density and time) for energy-aware operation within the tested domain. Quantitative energy projections (e.g., kWh·m⁻³) are not drawn from the present model outputs and require dedicated electro-energy

auditing under experimental validation. The model's generalization capability across operational conditions within this dataset suggests potential for transfer-learning extensions to related PFAS (e.g., PFOS, GenX), contingent on coverage of comparable operating ranges and external validation [47].

3.5. Limitations and future directions

Although FLAML demonstrated exceptional predictive accuracy and robustness when applied to the dataset curated by [30], several avenues remain promising for further exploration. First, future research could assess the adaptability of this modeling framework to broader operational parameter spaces and validate its performance using external datasets from different experimental setups. This would also include a wider spectrum of PFAS compounds, such as PFOS and co-contaminants, to ascertain the generalizability of insights across diverse treatment systems. Transfer learning methodologies hold particular promise in this regard, as they could efficiently leverage knowledge from the current model to accelerate predictive capabilities in related electrochemical systems.

Additionally, while the current study prioritized removal efficiency, extending the analytical framework towards integrated multi-objective optimization could significantly enhance practical applicability. Such an extension would consider critical performance indicators beyond contaminant removal, including energy consumption metrics, byproduct formation, and the computational trade-offs between optimization speed and final model performance. Furthermore, enhancing real-time adaptive process control represents another promising direction. Integrating FLAML with in situ sensing technologies would enable dynamic process optimization, facilitating immediate adjustments to operational parameters based on real-time feedback.

Finally, while SHAP analysis provided valuable initial insights, deeper interpretability could be achieved by developing hybrid modeling frameworks that combine FLAML with physics-based electrochemical models. Future methodological refinements could also include exploring the use of native categorical handling within tree-based models as a potential alternative to one-hot encoding. Collectively, these future research directions promise substantial advancements in predictive analytics, interpretability, and operational applicability, reinforcing the potential of AutoML as a transformative approach for environmental electrochemical remediation technologies.

4. Conclusions

This study demonstrates a reproducible AutoML-based workflow for integrating automated machine learning with electrochemical oxidation processes targeting persistent environmental contaminants. The FLAML-driven XGBoost model outperformed conventional manually optimized ML approaches, delivering significantly enhanced predictive accuracy and interpretability, as evidenced by rigorous benchmarking and statistical validation. The analysis identified electrolysis time and anode material as primary operational drivers, offering actionable insights grounded in fundamental electrochemical principles and validated through SHAP-based interpretability. Crucially, the automated FLAML framework substantially reduced computational demands (72 % reduction in tuning time), thereby enhancing the accessibility and reproducibility of advanced ML techniques for environmental practitioners. By demonstrating superior predictive power alongside scientifically robust mechanistic insights, this research directly addresses critical methodological limitations in the current literature. The containerized, version-pinned pipeline ensures reproducibility and transparency, promoting wider adoption and methodological standardization within environmental remediation. Moving forward, integrating the developed methodology with hybrid physics-based models, real-time adaptive process controls, and broader PFAS compound scenarios can further elevate electrochemical treatment efficacy and

versatility. Overall, this work firmly positions automated and interpretable machine learning as an essential pillar of environmental engineering innovation, advancing the global effort towards sustainable remediation of persistent environmental contaminants.

CRediT authorship contribution statement

Haitham Elnakar: Writing – review & editing, Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at [doi:10.1016/j.dwt.2025.101598](https://doi.org/10.1016/j.dwt.2025.101598).

Data Availability

The dataset used in this research has been cited in the methodology section.

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